

GAME DESIGN DOCUMENT · V1.3

Zombie Survival 2.5D Systems Design

Hero classes, weapon tiers, pet evolution, guardian system,
in-run upgrades, economy & progression. Built on the
existing Godot 4.5 codebase.

TABLE OF CONTENTS

1. Vision & Core Loop
2. Hero & Class System
3. Weapon System
4. Pet Companion System
5. Guardian System NEW
6. In-Run Upgrade System
7. Economy & Progression
8. Implementation Priority

1. Vision & Core Loop

A mobile-first 2.5D roguelite where each run is 5—15 minutes. Players choose a hero class, equip a weapon, pick a pet companion, and survive escalating waves of zombies. Between runs, they spend currency to unlock new heroes, upgrade equipment, and evolve pets.

Core Gameplay Loop

PREPARE

Select hero → equip
weapon → choose pet →
pick guardian → enter level

FIGHT

Survive waves → collect XP
→ pick in-run upgrades →
defeat boss every 5 waves

REWARD

Earn gold + gems + loot →
unlock/upgrade → repeat
with stronger loadout

Session Target

METRIC	TARGET	NOTES
Run length	5–15 min	Mobile attention span
Waves per level	10–20	Boss every 5 waves
Sessions per day	3–5	Energy system paces this
First unlock	< 3 runs	Hook before churn

2. Hero & Class System

Each hero has a distinct class identity with unique stats, one passive trait, two active skills, and one ultimate ability unlocked through hero-level progression.

2.1 Hero Roster

HERO	CLASS	ROLE	HP	SPEED	DMG	IDENTITY
Knight	Tank	Frontline	+4	+0.0	+1	High survivability, AoE taunt
Rogue	Assassin	DPS	+0	+1.0	+0	Speed & crit, hit-and-run
Mage	Caster	AoE	-2	+0.2	+2	Glass cannon, elemental AoE
Ranger NEW	Marksman	Ranged	+0	+0.3	+1	Long range, piercing shots
Paladin NEW	Support	Hybrid	+2	+0.0	+1	Heals, buffs pet & guardian

Current state: Knight, Rogue, and Mage exist in `heroes.json` with basic stats. Ranger and Paladin are proposed additions. Skill system is entirely new.

2.2 Class Skill Design

Each hero unlocks abilities through **Hero Level** (earned from XP across runs):

UNLOCK	HERO LEVEL	TYPE
Passive Trait	1 (default)	Always active
Skill 1	3	Active, cooldown-based
Skill 2	7	Active, cooldown-based
Ultimate	12	Charges from kills

Knight — Iron Bulwark



Iron Skin PASSIVE

Reduce all incoming damage by 1 (minimum 1 damage taken). Contact damage cooldown +0.3s.



Shield Bash ACTIVE · 8S CD

Slam forward, dealing 4 damage in a cone and knocking enemies back. Stuns bosses for 1.5s.



War Cry ACTIVE · 14S CD

Taunt all enemies within 8m for 3s. During taunt, Knight gains +30% move speed.



Bulwark ULTIMATE · 25 KILLS

Create an impenetrable shield dome (radius 5m) for 4s. All allies inside take 0 damage. Enemies touching the dome take 2 damage/s.

Rogue — Shadow Blade



Critical Edge PASSIVE

20% chance to deal 2× damage. Crit chance increases by 5% for each consecutive hit (resets on miss). Max 40%.



Shadow Dash ACTIVE · 5S CD

Dash forward 6m, passing through enemies. Next 3 attacks within 2s deal +50% damage.



Blade Storm ACTIVE · 12S CD

Throw 8 spinning blades in all directions, each dealing 2 damage. Blades pierce through enemies.



Phantom Strike ULTIMATE · 20 KILLS

Become invisible for 3s. Move speed +100%. First attack from stealth deals 5× damage and hits all enemies in a line.

Mage — Arcane Warden



Arcane Amplification PASSIVE

Projectiles that travel >70% of max range deal +30% damage. Explosion radius +15%.



Fireball ACTIVE · 6S CD

Launch a fireball that explodes on impact for 3 AoE damage (radius 3.5m). Leaves a burn zone for 2s dealing 1 damage/s.



Frost Nova ACTIVE · 10S CD

Freeze all enemies within 5m for 2s. Frozen enemies take +25% damage. Boss freeze duration halved.



Meteor Storm ULTIMATE · 30 KILLS

Call down 5 meteors over 3s across the screen, each dealing 6 AoE damage (radius 4m). Screen shake on impact.

Ranger — Hawk Eye (Proposed)



Eagle Sight PASSIVE

+20% projectile range. Enemies below 25% HP are highlighted and take +1 damage from all sources.



Piercing Shot ACTIVE · 7S CD

Fire a high-velocity arrow that pierces all enemies in a line, dealing 3 damage each. Range 1.5× normal.



Trap Field ACTIVE · 15S CD

Place 3 traps in a triangle pattern. Each trap explodes when an enemy walks over it, dealing 2 damage and slowing by 40% for 2s.



Arrow Rain ULTIMATE · 25 KILLS

Rain arrows over a large area (radius 8m) for 3s, dealing 1 damage per hit. 12 hits total, randomly distributed.

Paladin — Divine Light (Proposed)



Holy Aura PASSIVE

Pet companion and Guardian deal +20% damage. Player regenerates 1 HP every 20s.



Heal Pulse ACTIVE · 12S CD

Heal self for 3 HP. If at full HP, grant a shield that absorbs 2 damage (expires after 5s).



Smite ACTIVE · 8S CD

Strike the nearest enemy with holy light for 4 damage. If the target dies, the bolt chains to 2 nearby enemies for 2 damage each.



Divine Intervention ULTIMATE · 30 KILLS

Fully heal and become invulnerable for 2s. All enemies within 6m take 3 holy damage. Pet and Guardian attack speed doubled for 5s.

2.3 Hero Leveling

Heroes gain **Hero XP** after each run (win or lose). Hero XP is separate from in-run XP.

HERO LEVEL	XP REQUIRED	CUMULATIVE	UNLOCK
1	0	0	Base stats + Passive
2	100	100	+1 base stat point
3	200	300	Skill 1 unlocked
5	400	1,100	+2 base stat points
7	600	2,500	Skill 2 unlocked
10	1,000	5,500	Enhanced passive
12	1,500	8,500	Ultimate unlocked
15	2,000	14,500	Max level, all slots open

2.4 Implementation Notes

Data structure: Add to `heroes.json` per hero:

```
passive_id , skill_1_id , skill_2_id , ultimate_id , unlock_hero_levels : [1, 3, 7, 12]
```

New file: `data/skills.json` — all skill definitions with cooldowns, damage, radius, duration, VFX scene paths. Separate from hero data so skills can be referenced by the upgrade system too.

Player.gd refactor: Extract `SkillManager` component (~200 lines) that reads skill definitions, manages cooldowns, emits `skill_activated` signal. `Player.gd` calls

```
SkillManager.try_activate(slot_index) .
```

3. Weapon System

Weapons have rarity tiers, upgrade paths, and special effects. The weapon you bring into a run defines your primary attack pattern. Upgrades are persistent (between runs).

3.1 Rarity Tiers

TIER	COLOR	STAT MULTIPLIER	SPECIAL SLOTS	DROP RATE
Common	Gray	1.0×	0	50%
Uncommon	Green	1.15×	1	30%
Rare	Blue	1.35×	1	14%
Epic	Purple	1.6×	2	5%
Legendary	Gold	2.0×	2 + unique	1%

3.2 Weapon Types & Base Stats

WEAPON	TYPE	DMG	FIRE RATE	PROJ.	SPREAD	SPEED	RANGE	PATTERN
Basic Gun	Single	1	0.6s	1	0°	14	20	Straight line, reliable
Spread Shot	Shotgun	1	0.75s	3	24°	13	17	Fan pattern, close range
Rapid Blaster	Auto	1	0.32s	1	0°	16	18	High DPS, single target
Heavy Launcher	Heavy	3	0.9s	1	0°	11	23	Slow, massive per-hit
Chain Gun NEW	Beam	1	0.15s	1	0°	20	14	Ramps up DPS over time
Bouncer NEW	Ricochet	2	0.7s	1	0°	12	16	Bounces to 2 nearby enemies

3.3 Weapon Special Effects

Unlocked via **Special Slots** (determined by rarity). Each weapon can equip effects from a shared pool:

EFFECT	DESCRIPTION	CLASS SYNERGY
Burn	10% chance to ignite, 1 DMG/s for 3s	Mage +5% chance
Freeze	8% chance to slow enemy 40% for 2s	Mage +4% chance
Vampiric	Heal 1 HP per 15 kills	Paladin: per 10 kills
Piercing	Projectiles pass through 1 extra enemy	Ranger: 2 extra
Explosive	5% chance for AoE on hit (radius 2.5m)	Knight: +2% chance
Homing	Slight auto-aim toward nearest enemy	All classes

3.4 Weapon Upgrade System

Weapons level up using **Gold** + duplicate weapon drops (or universal upgrade materials):

LEVEL	GOLD COST	MATERIALS	STAT BOOST
1 → 2	100	1× duplicate	+10% damage
2 → 3	250	2× duplicate	+10% fire rate
3 → 4	500	3× duplicate	+10% damage
4 → 5	1,000	5× duplicate	Unlock weapon perk

Implementation: Add `rarity`, `level`, `special_effects` array, and `weapon_perk_id` to each weapon entry in `weapons.json`. `SaveManager` already has `inventory` — store weapon instances with unique IDs there.

4. Pet Companion System — Passive Support

Pets provide **passive buffs and economic bonuses** — they do not attack enemies directly. Active combat is the Guardian's job (Section 5). Pets follow the player visually and grant aura-style bonuses. They evolve through 3 stages, gaining stronger buffs at each evolution.

Key design rule: Pet = passive buff / economy / evolution system. Guardian = active combat ally. This separation keeps systems clean and avoids overlapping roles.

4.1 Pet Roster & Passive Buffs

PET	TYPE	BASE BUFF	AURA RANGE	SPECIAL
Drone	Offensive	Player damage +10%	–	Marks enemies near player, marked enemies take +1 DMG
Sprite	Speed	Player move speed +8%	–	Fire rate +5%. XP magnet range +20%
Wisp	Economy	Gold pickup +15%	–	Chance to find bonus gems on boss kill (+10%)
Golem NEW	Defense	Player max HP +2, damage taken –1	–	Guardian HP +15% (synergy)
Fairy NEW	Healing	Regenerate 1 HP every 15s	–	Skill cooldowns –10%

Current state: `PetCompanion.gd` currently has active attack logic (`_find_nearest_enemy()`, projectile firing). This needs refactoring — remove attack code, replace with buff/aura system that applies stat modifiers to the player and guardian.

4.2 Pet Evolution

Pets evolve using **Pet Shards** (dropped from bosses or earned from daily rewards). Each stage upgrades the passive buff and adds a new bonus:

STAGE	SHARDS	GOLD	BUFF CHANGE
Stage 1 — Base	0	0	Base passive buff only
Stage 2 — Evolved	15	500	+50% buff strength, new visual, gains secondary passive
Stage 3 — Ascended	40	2,000	+100% buff strength, final visual, gains triggered ability

Evolution Example — Drone

STAGE	NAME	PRIMARY BUFF	BONUS
1	Drone	Player damage +10%	Marks nearby enemies (+1 DMG from player)
2	Combat Drone	Player damage +15%	+ Mark also applies to Guardian targets
3	War Machine	Player damage +20%	+ Every 20 kills: AoE shockwave (3 DMG, 5m radius)

4.3 Pet Equipment

Each pet has **1 accessory slot**. Accessories enhance the pet's buff or add new passive effects:

ACCESSORY	EFFECT	RARITY
Buff Amplifier	Pet's base buff +25%	Common
Fortune Charm	Gold pickup +10% (stacks with pet buff)	Uncommon
Shield Amulet	Absorb 1 hit every 30s	Rare
Lifeline Core	Regen 1 HP every 20s (in addition to pet buff)	Epic
Synergy Crystal	Guardian damage +20%, Guardian attack speed +15%	Legendary

Implementation: Refactor `PetCompanion.gd` : remove `_find_nearest_enemy()` and attack logic. Add `BuffProvider` component that emits stat modifiers. `Player.gd` reads pet buffs via `pet.get_active_buffs()` . Add `evolution_stage` , `evolution_shards` , `accessory_id` to `SaveManager`. New file: `data/pet_evolutions.json` with per-stage buff definitions.

5. Guardian System — Skill-Based Combat Support

NEW SYSTEM

Guardians are **active support followers** that travel with the player and use cooldown-based skills to help survive. They do not use guns like the player — they use melee, AoE, and utility skills on long cooldowns. Only one Guardian can be active per run.

Core fantasy: Player bị bao vây → Guardian dùng skill mở đường → Player tiếp tục kite.
Guardian không giết quái thay player. Guardian tạo cơ hội để player sống sót.

5.1 Core Direction

SYSTEM	ROLE	COMBAT STYLE
Player	Main DPS (gun combat)	Directly controlled, continuous fire
Pet	Passive buff / economy	Does not attack; provides auras & stat bonuses
Guardian	Skill-based combat support	Cooldown skills — CC, burst, heal, utility

This trio is clean and easy to expand. Each system has a distinct identity.

5.2 Guardian Design Rules

✓ GUARDIAN SHOULD

- Follow player at short distance
- Use skills when conditions are met
- Have clear, long cooldowns
- Deal moderate damage (~20—30% of player DPS)
- Provide crowd control, heal, or utility
- Not spam attacks every second

⊘ GUARDIAN SHOULD NOT

- Shoot continuously like player
- Become a second pet
- Scan enemies every frame
- Exceed 50% of player DPS
- Replace hero skills
- Use projectiles (MVP)

5.3 MVP Guardian — Bruiser Guard

MVP scope: Ship with one Guardian type only — a melee Bruiser with cooldown-based skills. Easiest to implement (no projectiles), feels good immediately, and is clearly distinct from pet & player.

GUARDIAN	TYPE	HP	COMBAT STYLE	SKILLS	FANTASY
Bruiser Guard	Melee CC	—	Cooldown skills (no auto-attack)	3	Bodyguard with crude weapon (club, axe, short sword, shield) — pushes zombies away

Skill 1 — Slam Knockback **MOST IMPORTANT**

Purpose: mở đường khi player bị bao vây.

PROPERTY	VALUE
Cooldown	5s
Range	2.5m radius around guard
Damage	1–2
Effect	Push zombies outward + light hit stun (if available) + camera shake (light) + shockwave / dust VFX

Skill 2 — Short Cleave

Purpose: đánh quái gần guard, tạo cảm giác guard đang chiến đấu.

PROPERTY	VALUE
Cooldown	2.5s
Range	Very short, frontal cone
Damage	1
Effect	Hits multiple enemies in small fan shape + hit spark VFX + small slash effect

Skill 3 — Emergency Heal

Purpose: cứu player khi máu thấp — great for retention ("may quá guard cứu mình!").

PROPERTY	VALUE
Cooldown	12–15s
Trigger	Auto when player HP < 40%
Heal	1–2 HP
Effect	Green pulse / light flash on player. Cannot be spammed.

For MVP, Bruiser Guard is **invulnerable** (no HP system needed). Lasts until run ends. No projectiles — all melee/AoE. Reuses existing VFX.

MVP ACCEPTANCE CRITERIA

- Bruiser Guard spawns during gameplay
- Follows player at short distance
- Uses skills on cooldown when conditions are met
- Helps control enemies (knockback, cleave)
- Improves survival but does not replace player
- Gameplay loop remains stable

5.4 Five Guardian Kits (Full Roster)

All guardians use cooldown-based skills. Each has 3 skills, strengths, and weaknesses.

5.4.1 Bruiser Guard MVP

Role: melee crowd control • **Best for:** new players, survival builds

SKILL	COOLDOWN	DESCRIPTION
Slam Knockback	5s	Push zombies away, small damage
Short Cleave	2.5s	Frontal arc, multiple enemies, low damage
Emergency Heal	12–15s	Auto-heal when player HP < 40%

✓ Easy to understand, no projectiles, great emergency saves

⚠ Low damage, useless against distant enemies

5.4.2 Grenadier Guard

Role: AoE burst support • **Best for:** clearing zombie hordes

SKILL	COOLDOWN	DESCRIPTION
Throw Grenade	6–8s	Lob grenade at nearest enemy cluster. Small explosion radius.
Smoke Push	10s	Create short smoke zone — enemies inside are slowed or pushed.
Blast Finish	Long	If many enemies surround player, trigger a small explosion.

✓ Very satisfying, excellent when swarmed

⚠ Long cooldowns, low value against few enemies

5.4.3 Medic Guard

Role: defensive support • **Best for:** weaker players, tank builds

SKILL	COOLDOWN	DESCRIPTION
Emergency Heal	Long	Heal player when HP is low
Cleanse Pulse	Medium	Light push on enemies near player — no heavy damage
Fortify	15s	Reduce damage player receives for 3–4s

✓ Strong retention — player feels "saved by guard". Great monetization hook.

⚠ Doesn't increase clear speed. Can make game too easy if heal is too strong.

5.4.4 Blade Guard

Role: short-range burst DPS • **Best for:** aggressive players, fast combat

SKILL	COOLDOWN	DESCRIPTION
Dash Slash	Medium	Lunge to nearest enemy, slash, return to player
Whirl Cut	6s	360° slash around guard, hits multiple nearby enemies
Bleed Strike	Medium	Apply light bleed DoT for a few seconds (no stacking)

- ✓ Looks action-packed, decent damage while still melee
- ⚠ Can overlap with player if too strong. Needs careful pathfinding.

5.4.5 Barrier Guard

Role: protection / space control • **Best for:** boss waves, defensive play

SKILL	COOLDOWN	DESCRIPTION
Shield Push	5s	Push enemies in front of player. Low damage.
Barrier Wall	Long	Create temporary barrier (2—3s). Enemies touching it are slowed/knocked.
Guard Stance	Passive/hybrid	When player HP is low, guard stays closer and reduces incoming damage.

- ✓ Creates tactical kiting around barriers. Very different from pet/player.
- ⚠ Low damage. Needs careful tuning to not slow the game down.

5.5 Guard AI Design

5.5.1 State Machine

Guardian uses a simple 4-state loop — no complex state machine needed for MVP:

FOLLOW Stay near player with slight offset. No pathfinding needed — simple movement toward player position.	EVALUATESKILL Every 0.2s: find alive enemies near player/guard, check cooldowns, pick best skill.	CASTSKILL Play animation/effect, apply damage/knockback/heal, reset cooldown timer.	COOLDOWN Skills recharging. Guard continues following player. Returns to EvaluateSkill on next scan.
---	---	---	--

5.5.2 Targeting Rules

SETTING	VALUE
Scan interval	0.2s (not per frame)
Target filter	Skip dead enemies
Priority	Enemies nearest to player first

5.5.3 Skill Priority Order

#	CONDITION	ACTION
1	Player HP < 40% AND heal cooldown ready	Emergency Heal
2	2+ enemies within radius near player AND slam cooldown ready	Slam Knockback
3	Enemy within short range AND cleave cooldown ready	Short Cleave
4	No conditions met	Continue following player

5.6 Skill Trigger Logic (Reference)

Emergency Heal

```
if player.hp < player.max_hp * 0.4 AND heal_cooldown_ready:  
→ heal player 1—2 HP, play green pulse effect, reset cooldown 12—15s
```

Slam Knockback

```
if count_enemies_in_radius(guard.position, 2.5) >= 2 AND slam_cooldown_ready:  
→ damage 1—2, knockback outward, optional short stun, shockwave VFX, reset cooldown 5s
```

Short Cleave

```
if nearest_enemy_distance < cleave_range AND cleave_cooldown_ready:  
→ damage enemies in frontal cone, hit spark, small slash VFX, reset cooldown 2.5s
```

5.7 Balance Guide

DPS BUDGET

Guardian should contribute approximately **20—30% of player DPS**. Never exceed 50% — player must remain the main damage dealer and feel important.

SKILL	COOLDOWN	DAMAGE
Short Cleave	2.5s	1
Slam Knockback	5s	1–2
Emergency Heal	12–15s	0 (heals 1–2 HP)

These values are starting points. Tune after playtesting — guard damage should never feel like "I can stop shooting and let guard kill them."

5.8 Monetization Direction

Guardian is an ideal **rewarded-ad feature** — player hires a helpful ally in exchange for watching an ad.

MVP

ACTION	RESULT
Watch Ad / Hire Guard	Spawn Bruiser Guard for current run
No Ad SDK yet?	Debug "Hire Guard" button as placeholder

Future Monetization

- Watch ad → hire guardian for one run
- Spend gems → hire without ad
- Guardian skins (cosmetic, gems-only)
- Guardian upgrades (permanent level-ups)
- Guardian rarity system
- Duration boost (extend guardian time via ad)
- Revive guardian mid-run via ad (post-MVP when HP exists)

Golden rule: Guardian giúp dễ chơi hơn, nhưng **không bắt buộc để thắng**. Runs must be completable without a Guardian.

5.9 Guardian Progression

LEVEL	COST	BONUS
1	Free (hired via ad or debug)	Base stats
2	Gold + Guardian Stones	+20% skill damage
3	Guardian Stones	Unlock passive perk
4	Gold + Stones	−15% cooldowns
5	Rare material	Ultimate skin + enhanced skill effect

5.10 System Identity

SYSTEM	IDENTITY	FEELING
Player	Gun combat	"I'm the main fighter"
Pet	Passive support	"My pet boosts me"
Guardian	Cooldown skill support	"My bodyguard saved me!"

This trio is clean, distinct, and easy to expand post-launch.

5.11 Implementation Notes

New data: `data/guardians.json`

Fields: `id`, `display_name`, `type`, `skills[]` (each with `name`, `cooldown`, `damage`, `range`, `trigger_condition`), `follow_distance`, `model_scene_path`, `hire_duration`, `rarity`, `unlock_condition`

New scene: `scenes/entities/guardians/bruiser_guard.tscn`

New script: `scripts/entities/guardian/BruiserGuard.gd`

State machine: Follow → EvaluateSkill (every 0.2s) → CastSkill → Cooldown → Follow

VFX reuse: hit spark, explosion, camera shake, dust — no new VFX needed for MVP

Cleanup: `queue_free()` on game over or scene exit

GameManager additions:

```
selected_guardian_id , active_guardian_ref , request_hire_guard(guard_id) ,  
on_rewarded_ad_completed_hire_guard(guard_id)
```

6. In-Run Upgrade System

Between waves, players choose one upgrade from a randomized selection. Upgrades stack during the run but reset when the run ends. The system should feel like meaningful choices, not auto-pilot.

6.1 Upgrade Pool

Generic Upgrades (Available to All Heroes)

UPGRADE	EFFECT PER STACK	MAX STACKS	RARITY
Quick Trigger	Fire rate +12%	5	Common
Heavy Rounds	Projectile damage +1	4	Common
Swift Feet	Move speed +10%	5	Common
Vitality	Max HP +2	5	Common
Multi-Shot	+1 projectile per attack	2	Rare
Explosive Rounds	Projectiles explode on hit (3m AoE)	1	Rare
Magnetic Field	XP pickup range +40%	3	Common
Ricochet	Projectiles bounce to 1 nearby enemy	2	Epic
Second Wind	Revive once with 30% HP when killed	1	Legendary

Hero-Specific Upgrades

HERO	UPGRADE	EFFECT	RARITY
Knight	Thorns Armor	Reflect 1 damage to enemies on contact	Uncommon
Knight	Fortress	+3 HP, but move speed -15%	Rare
Rogue	Poison Blade	Crits apply poison (1 DMG/s for 3s)	Uncommon
Rogue	Shadow Clone	Dash leaves a decoy that attacks for 3s	Epic
Mage	Mana Overflow	Skill cooldowns -20%	Uncommon
Mage	Chain Lightning	Projectiles chain to 2 enemies for 50% damage	Rare
Ranger	Eagle Eye	Projectile range +30%, crit rate +10%	Uncommon
Paladin	Divine Shield	Absorb next 3 hits (refreshes every 30s)	Rare

6.2 Upgrade Selection Rules

- 3 choices shown per wave clear (can re-roll once using gems)
- At least 1 hero-specific upgrade offered if available
- Rarity weighted: Common 55%, Uncommon 25%, Rare 14%, Epic 5%, Legendary 1%
- Already-maxed upgrades removed from pool
- Boss wave rewards offer 1 rarity tier higher minimum

Implementation: Current upgrade system in GameManager has 6 hardcoded IDs. Refactor to data-driven: `data/upgrades.json` with `hero_restriction`, `max_stacks`, `rarity`, `stat_effects` array. `begin_upgrade_selection()` draws from the weighted pool. Add `re_roll_upgrade_selection()` that costs 5 gems.

7. Economy & Progression

Two currencies (Gold, Gems) plus collectible materials. The economy must feel generous enough to retain F2P players while providing clear value for spenders.

7.1 Currency Design

CURRENCY	EARN RATE	PRIMARY SINKS	PURCHASABLE
Gold (soft)	50—200 per run, scales with wave reached	Weapon upgrades, pet evolution, guardian levels	No (earned only)
Gems (hard)	5—15 per run, daily rewards, achievements	Re-rolls, hero unlocks, premium cosmetics, energy refill	Yes (IAP)
Pet Shards	1—3 per boss kill, daily login	Pet evolution	Via gem conversion
Guardian Stones	2—5 per level clear	Guardian upgrades	Via gem conversion
Hero XP	Based on run performance (kills, waves, time)	Hero leveling	No

7.2 Energy System

PARAMETER	VALUE
Max energy	5
Energy per run	1
Regen rate	1 per 30 min
Full refill cost	20 gems
Ad refill	+1 energy (max 3 per day)

7.3 Unlock Pacing

MILESTONE	RUNS	WHAT UNLOCKS
First play	1	Knight + Basic Gun + Drone. Tutorial run.
First upgrade	2	Enough gold for first weapon level-up
Second hero	5	Rogue unlocked (gold or achievement)
First pet evolution	10	15 pet shards collected
Third hero	15	Mage unlocked (gold)
Guardian system	8	Unlocked after clearing Level 2. First guardian free.
Second guardian	20	Enough guardian stones
Ranger hero	30	Gems or achievement
Paladin hero	50	Gems only (premium feel)

7.4 Monetization Model — Hybrid (F2P + IAP + Ads)

REVENUE STREAM	DETAILS	EST. REVENUE SHARE
Gem Packs	\$0.99 (60 gems), \$4.99 (350 gems), \$9.99 (800 gems), \$19.99 (1,800 gems)	40%
Starter Pack	\$2.99 one-time: 200 gems + Rogue + Rare weapon. Best value.	15%
Season Pass	\$4.99/month: daily gem bonus, exclusive skins, +1 max energy	25%
Rewarded Ads	Double run rewards, free energy, free re-roll. Opt-in only.	20%

Golden rule: No gameplay-critical content behind a paywall. Every hero, weapon, pet, and guardian is earnable through gameplay. Gems accelerate — never gate. Cosmetics can be gems-exclusive (skins, effects, titles).

7.5 Daily / Weekly Engagement

FEATURE	REWARD	RESET
Daily login	Gold (Day 1—6), Gems (Day 7)	Daily
Daily quest (3×)	"Kill 50 enemies", "Clear 3 waves"...	Daily
Weekly challenge	"Clear Level 3 with Mage" → Pet Shards	Weekly
Achievement	One-time gem + cosmetic rewards	Permanent

8. Implementation Priority

Ordered by impact on gameplay loop. Each phase should be playable and testable before starting the next.

PHASE	SYSTEM	EST. EFFORT	DEPENDENCIES
0 — Bug Fixes	HP double-apply, dead-enemy targeting, camera shake, tmp file, TEST flag	1 day	None
1A — Data Layer	skills.json, upgrades.json, guardians.json, pet_evolutions.json schema	2–3 days	Phase 0
1B — Code Health	Split player.gd (WeaponVisuals, SkillManager), split home_screen.gd	3–4 days	Phase 0
2 — Hero Skills	SkillManager, skill UI (cooldown indicators), 3 hero skill sets	1–2 weeks	Phase 1A + 1B
3 — Weapon Tiers	Rarity system, weapon upgrades, special effects, loot drops	1 week	Phase 1A
4 — Pet Refactor	Remove pet attack code, add buff/aura system, evolution UI, accessories	1 week	Phase 1A + 1B
5A — Guardian MVP	Bruiser Guard: follow player, Slam + Cleave + Emergency Heal skills, invulnerable	3–4 days	Phase 1A
5B — Guardian Full	4 more guardian types, HP system, destruction/respawn, synergies	1–1.5 weeks	Phase 5A
6 — Economy	Energy system, daily rewards, quests, shop UI, gem store	1–2 weeks	Phase 3 + 4
7 — Monetization	IAP integration (Google/Apple), ad SDK, season pass	1 week	Phase 6
8 — New Heroes	Ranger + Paladin with full skill sets and models	1 week	Phase 2

Total estimated timeline: 8—12 weeks for a solo developer, 4—6 weeks for a 2-person team. Phase 0 + 1 should be done first regardless — they're prerequisites for everything else and they fix real bugs in the current build.

This document is a living draft. All numbers (damage, costs, timing) should be tuned through playtesting. Start with these values, then adjust based on average run length, death rate, and progression speed.